

Saber Jafarpour

INFORMATION	Research Assistant Professor Department of Computer Science University of Colorado Boulder, ECES 118E	Email: saber.jafarpour@colorado.edu Google Scholar: Saber Jafarpour Orcid ID: 0000-0002-7614-2940 Webpage: saberjafarpour.github.io
ACADEMIC POSITIONS	University of Colorado Boulder Department of Computer Science <i>Research Assistant Professor</i>	2023–present
	Georgia Institute of Technology Institute for Robotics and Intelligent Machines <i>Postdoctoral Research Fellow (Advisor: Samuel Coogan)</i>	2021–2023
	University of California Santa Barbara Center of Control, Dynamical Systems, and Computation <i>Postdoctoral Research Fellow (Advisor: Francesco Bullo)</i>	2016–2021
EDUCATION	Queen’s University, Canada Department of Mathematics and Statistics <i>Ph.D. in Applied Mathematics (Advisor: Andrew D. Lewis)</i> Dissertation: On the Role of Regularity in Mathematical Control Theory	2011–2016
	Shiraz University, Iran Department of Mechanical Engineering <i>M.Sc. in Applied Mechanics (Advisor: Mojtaba Mahzoon)</i> Dissertation: Investigating Nonholonomic Constraints in Robotic Systems	2008–2011
	Shiraz University, Iran Department of Mechanical Engineering <i>B.Sc. in Mechanical Engineering</i>	2004–2008
RESEARCH INTERESTS	My research interests lie at the broad intersection of machine learning, autonomy, and control theory. I develop mathematically rigorous and computationally efficient tools for the safety, learning, and control of autonomous systems, with applications in robotics and cyber-physical systems. I have been conducting research in the following areas: • Safety assurance for learning-enabled systems • Uncertainty quantification in autonomous systems • Resilience of large-scale multi-agent systems • Geometric control of nonlinear systems	
UNDER REVIEW PAPERS	[U4] A. Alavi, M. Zamani, and S. Jafarpour. Scalable neural safety certification via monotonicity, 2026a. Submitted [U3] F. Mafi, S. Jafarpour, and M. Pirani. A separation principle for adversarial attacks and defense of graph neural networks, 2026. Submitted	

- [U2] D. Ajeyemi, A. Colot, S. Dhople, E. Dall’Anese, and **S. Jafarpour**. Reachability analysis for power systems with heterogeneous resources via jordan transformation, 2026. URL <https://arxiv.org/abs/2607.05599>. Submitted
- [U1] F. Galarza-Jimenez, M. Zamani, and **S. Jafarpour**. Trajectory-based safety of monotone systems: Verification and control synthesis, 2026. URL <https://arxiv.org/abs/2605.16660>. Submitted
- [J27] Z. Liu, **S. Jafarpour**, and Y. Chen. Safety verification of nonlinear stochastic systems via probabilistic tube. *IEEE Transactions on Automatic Control*, 2026b. DOI: [10.1109/TAC.2026.3670756](https://arxiv.org/abs/2503.03328). URL <https://arxiv.org/abs/2503.03328>
- [J26] Z. Liu, **S. Jafarpour**, and Y. Chen. Probabilistic reachability of discrete-time nonlinear stochastic systems. *Automatica*, 190:113063, 2026a. DOI: [10.1016/j.automatica.2026.113063](https://www.arxiv.org/abs/2409.09334). URL <https://www.arxiv.org/abs/2409.09334>
- [J25] A. Alavi, A. Nadali, M. Zamani, and **S. Jafarpour**. Neural barrier certificates for monotone systems. *IEEE Control Systems Letters*, 9:1496–1501, 2025. DOI: [10.1109/LCSYS.2025.3579779](https://arxiv.org/abs/2503.03328). **(Finalist for Hybrid Systems Outstanding Student Paper Award)**
- [J24] S. Jafarpour, Z. Liu, and Y. Chen. Probabilistic reachability analysis of stochastic control systems. *IEEE Transactions on Automatic Control*, 70(11):7080–7094, 2025. DOI: [10.1109/TAC.2025.3566983](https://arxiv.org/abs/2503.03328)
- [J23] Z. Liu, **S. Jafarpour**, and Y. Chen. Safety verification of stochastic systems: A set-erosion approach. *IEEE Control Systems Letters*, 8:2859–2864, 2024. DOI: [10.1109/LCSYS.2024.3518394](https://arxiv.org/abs/2503.03328)
- [J22] A. Davydov*, **S. Jafarpour***, A. V. Proskurnikov, and F. Bullo. Non-Euclidean monotone operator theory and applications. *Journal of Machine Learning Research*, 25(307):1–33, Dec. 2024. URL <https://www.jmlr.org/papers/volume25/23-0805/23-0805.pdf>
- [J21] **S. Jafarpour***, A. Harapanahalli*, and S. Coogan. Efficient interaction-aware interval analysis of neural network feedback loops. *IEEE Transactions on Automatic Control*, 69(12):8706–8721, 2024. DOI: [10.1109/TAC.2024.3420968](https://arxiv.org/abs/2503.03328)
- [J20] **S. Jafarpour** and S. Coogan. Monotonicity and contraction on polyhedral cones. *IEEE Transactions on Automatic Control*, 70(2):1200–1207, 2025. DOI: [10.1109/TAC.2024.3454258](https://arxiv.org/abs/2503.03328)
- [J19] M. Pirani* and **S. Jafarpour***. Network critical slowing down: Data-driven detection of critical transitions in nonlinear networks. *IEEE Transactions on Control of Network Systems*, 11(2):573–585, 2024. DOI: [10.1109/TCNS.2023.3332730](https://arxiv.org/abs/2503.03328)
- [J18] **S. Jafarpour**, A. Davydov, and F. Bullo. Non-Euclidean contraction theory for monotone and positive systems. *IEEE Transactions on Automatic Control*, 68(9):5653–5660, 2023. DOI: [10.1109/TAC.2022.3224094](https://arxiv.org/abs/2503.03328)
- [J17] A. Davydov, **S. Jafarpour**, and F. Bullo. Non-Euclidean contraction theory for robust nonlinear stability. *IEEE Transactions on Automatic Control*, 67(12):6667–6681, 2022. DOI: [10.1109/TAC.2022.3183966](https://arxiv.org/abs/2503.03328)
- [J16] **S. Jafarpour**, P. Cisneros-Velarde, and F. Bullo. Weak and semi-contraction for network systems and diffusively-coupled oscillators. *IEEE Transactions on Automatic Control*, 67(3):1285–1300, 2022a. DOI: [10.1109/TAC.2021.3073096](https://arxiv.org/abs/2503.03328)
- [J15] **S. Jafarpour**, V. Purba, S. V. Dhople, B. Johnson, and F. Bullo. Singular perturbation and small-signal stability for inverter networks. *IEEE Transactions on Control of Network Systems*, 9(2):979–992, 2022c. DOI: [10.1109/TCNS.2021.3084444](https://arxiv.org/abs/2503.03328)
- [J14] P. Cisneros-Velarde, **S. Jafarpour**, and F. Bullo. A contraction analysis of primal-dual dynamics in distributed and time-varying implementations. *IEEE Transactions on Automatic Control*, 67(7):3560–3566, 2022. DOI: [10.1109/TAC.2021.3103865](https://arxiv.org/abs/2503.03328)
- [J13] K. D. Smith, **S. Jafarpour**, and F. Bullo. Transient stability of droop-controlled inverter networks with operating constraints. *IEEE Transactions on Automatic Control*, 67(2):633–645, 2022a. DOI: [10.1109/TAC.2021.3053552](https://arxiv.org/abs/2503.03328)
- [J12] K. D. Smith, **S. Jafarpour**, A. Swami, and F. Bullo. Topology inference with multivariate cumulants: The Möbius inference algorithm. *IEEE/ACM Transactions on Networking*, 30(5):2102–2116, 2022b. DOI: [10.1109/TNET.2022.3164336](https://arxiv.org/abs/2503.03328)

- [J11] **S. Jafarpour***, A. Davydov*, A. V. Proskurnikov, and F. Bullo. Robust implicit networks via non-Euclidean contractions. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 34, pages 9857–9868, Dec. 2021. URL <https://openreview.net/forum?id=SwfsoPuGYku>
- [J10] **S. Jafarpour**, E. Y. Huang, K. D. Smith, and F. Bullo. Flow and elastic networks on the n -torus: Geometry, analysis and computation. *SIAM Review (Research Spotlight)*, 64(1):59–104, 2021. DOI: [10.1137/18M1242056](https://doi.org/10.1137/18M1242056)
- [J9] A. Silva, F. Kocayusufoglu, **S. Jafarpour**, A. Swami, F. Bullo, and A. K. Singh. Combining physics and machine learning for network flow estimation. In *International Conference on Learning Representations*, Online, May 2021. URL <https://openreview.net/forum?id=10V53bErniB>
- [J8] X. Duan, **S. Jafarpour**, and F. Bullo. Graph-theoretic stability conditions for Metzler matrices and monotone systems. *SIAM Journal on Control and Optimization*, 59(5):3447–3471, 2021. DOI: [10.1137/20M131802X](https://doi.org/10.1137/20M131802X)
- [J7] **S. Jafarpour**. On small-time local controllability. *SIAM Journal on Control and Optimization*, 58(1):425–446, 2020. DOI: [10.1137/16M1068797](https://doi.org/10.1137/16M1068797)
- [J6] V. Purba, B. Johnson, **S. Jafarpour**, F. Bullo, and S. V. Dhople. Dynamic aggregation of grid-tied three-phase inverters. *IEEE Transactions on Power Systems*, 35(2):1520–1530, 2020. DOI: [10.1109/TPWRS.2019.2942292](https://doi.org/10.1109/TPWRS.2019.2942292)
- [J5] V. Purba, B. Johnson, M. Rodriguez, **S. Jafarpour**, F. Bullo, and S. V. Dhople. Reduced-order aggregate model for parallel-connected single-phase inverters. *IEEE Transactions on Energy Conversion*, 34(2):824–837, 2019. DOI: [10.1109/TEC.2018.2881710](https://doi.org/10.1109/TEC.2018.2881710)
- [J4] **S. Jafarpour**, E. Y. Huang, and F. Bullo. Synchronization of Kuramoto oscillators: Inverse Taylor expansions. *SIAM Journal on Control and Optimization*, 57(5):3388–3412, 2019. DOI: [10.1137/18M1216262](https://doi.org/10.1137/18M1216262)
- [J3] **S. Jafarpour** and F. Bullo. Synchronization of Kuramoto oscillators via cutset projections. *IEEE Transactions on Automatic Control*, 64(7):2830–2844, 2019. DOI: [10.1109/TAC.2018.2876786](https://doi.org/10.1109/TAC.2018.2876786)
- [J2] M. George, **S. Jafarpour**, and F. Bullo. Markov chains with maximum entropy for robotic surveillance. *IEEE Transactions on Automatic Control*, 64(4):1566–1580, 2019. DOI: [10.1109/TAC.2018.2844120](https://doi.org/10.1109/TAC.2018.2844120)
- [J1] **S. Jafarpour** and A. D. Lewis. Locally convex topologies and control theory. *Mathematics of Control, Signals and Systems*, 28(4):1–29, 2016b. DOI: [10.1007/s00498-016-0179-0](https://doi.org/10.1007/s00498-016-0179-0)

REFEREED
CONFERENCE
PAPERS

- [C19] A. Alavi, M. Zamani, and **S. Jafarpour**. Data-driven compositional safety verification of interconnected monotone systems. In *65th IEEE Conference on Decision and Control*, Honolulu, Hawaii, 2026b
- [C18] A. Makdesi, M. Zamani, and **S. Jafarpour**. Monotone embedding for hierarchical control of linear systems with disturbances. In *65th IEEE Conference on Decision and Control*, Honolulu, Hawaii, 2026b
- [C17] A. Makdesi, M. Zamani, and **S. Jafarpour**. Data-driven controlled invariance via monotone embeddings. In *23rd IFAC World Congress*, Busan, Korea, 2026a
- [C16] F. Galarza-Jimenez, M. Zamani, and **S. Jafarpour**. Trajectory-based barrier certificates for monotone systems. In *64th IEEE Conference on Decision and Control*, pages 5831–5836, 2025
- [C15] D. Ajeyemi, **S. Jafarpour**, and E. Dall’Anese. Neural network-assisted interval reachability for systems with control barrier function-based safe controllers. In *Proceedings of The 5th Annual Learning for Dynamics and Control Conference*, Proceedings of Machine Learning Research. PMLR, 6–8 Jun 2025. URL <https://arxiv.org/abs/2504.08249>. To appear
- [C14] A. Harapanahalli, **Jafarpour, S.**, and S. Coogan. immrax: A parallelizable and differentiable toolbox for interval analysis and mixed monotone reachability in JAX. *IFAC-PapersOnLine*, 58(11):75–80, 2024. DOI: [10.1016/j.ifacol.2024.07.428](https://doi.org/10.1016/j.ifacol.2024.07.428). 8th IFAC Conference on Analysis and Design of Hybrid Systems ADHS 2024
- [C13] **S. Jafarpour** and S. Coogan. A contracting dynamical system perspective toward interval markov decision processes. In *IEEE Conf. on Decision and Control*, pages 2918–2924, Marina Bay Sands, Singapore, Dec. 2023. DOI: [10.1109/CDC49753.2023.10383575](https://doi.org/10.1109/CDC49753.2023.10383575)

- [C12] A. Harapanahalli, **S. Jafarpour**, and S. Coogan. Contraction-guided adaptive partitioning for reachability analysis of neural network controlled systems. In *IEEE Conf. on Decision and Control*, pages 6044–6051, Marina Bay Sands, Singapore, Dec. 2023b. DOI: [10.1109/CDC49753.2023.10383360](https://doi.org/10.1109/CDC49753.2023.10383360)
- [C11] **Jafarpour, S.**, A. Harapanahalli, and S. Coogan. Interval reachability of nonlinear dynamical systems with neural network controllers. In *Proceedings of The 5th Annual Learning for Dynamics and Control Conference*, volume 211 of *Proceedings of Machine Learning Research*, pages 12–25. PMLR, 2023. URL <https://proceedings.mlr.press/v211/jafarpour23a.html>
- [C10] A. Harapanahalli, **Jafarpour, S.**, and S. Coogan. A toolbox for fast interval arithmetic in numpy with an application to formal verification of neural network controlled system. In *ICML workshop on Formal Verification of Machine Learning (WFVML 2023)*, 2023a. URL <https://arxiv.org/abs/2306.15340>
- [C9] **S. Jafarpour**, A. Davydov, M. Abate, F. Bullo, and S. Coogan. Robust training and verification of implicit neural networks: A non-Euclidean contractive approach. In *ICML Workshop on Formal Verification of Machine Learning*, July 2022b. DOI: [10.48550/arXiv.2208.03889](https://doi.org/10.48550/arXiv.2208.03889)
- [C8] **S. Jafarpour***, M. Abate*, A. Davydov*, F. Bullo, and S. Coogan. Robustness certificates for implicit neural networks: A mixed monotone contractive approach. In *Learning for Dynamics and Control Conf.*, volume 168, pages 917–930, June 2022. URL <https://proceedings.mlr.press/v168/jafarpour22a>. (Oral Presentation: Top 10 percent of submitted papers)
- [C7] A. Davydov*, **S. Jafarpour***, M. Abate, F. Bullo, and S. Coogan. Comparative analysis of interval reachability for robust implicit and feedforward neural networks. In *2022 IEEE 61st Conference on Decision and Control (CDC)*, pages 2073–2078, 2022. DOI: [10.1109/CDC51059.2022.9993217](https://doi.org/10.1109/CDC51059.2022.9993217)
- [C6] A. Davydov*, **S. Jafarpour***, A. V. Proskurnikov, and F. Bullo. Non-Euclidean monotone operator theory with applications to recurrent neural networks. In *IEEE Conf. on Decision and Control*, Cancún, México, Dec. 2022. DOI: [10.1109/CDC51059.2022.9993197](https://doi.org/10.1109/CDC51059.2022.9993197)
- [C5] **S. Jafarpour** and S. Coogan. Resilience of input metering in dynamic flow networks. In *American Control Conference*, pages 126–131, Atlanta, USA, June 2022. DOI: [10.23919/ACC53348.2022.9867237](https://doi.org/10.23919/ACC53348.2022.9867237)
- [C4] F. Bullo, P. Cisneros-Velarde, A. Davydov, and **S. Jafarpour**. From contraction theory to fixed point algorithms on Riemannian and non-Euclidean spaces. In *IEEE Conf. on Decision and Control*, Austin, USA, Dec. 2021. DOI: [10.1109/CDC45484.2021.9682883](https://doi.org/10.1109/CDC45484.2021.9682883). Invited Tutorial Session
- [C3] E. Y. Huang, **S. Jafarpour**, and F. Bullo. Synchronization of coupled oscillators: The Taylor expansion of the inverse Kuramoto map. In *IEEE Conf. on Decision and Control*, pages 5340–5345, Miami, USA, Dec. 2018. DOI: [10.1109/CDC.2018.8619559](https://doi.org/10.1109/CDC.2018.8619559)
- [C2] **S. Jafarpour** and A. D. Lewis. The classical and tautological orbit theorems. In *22nd International Symposium on Mathematical Theory of Networks and Systems*, July 2016a
- [C1] **S. Jafarpour** and A. D. Lewis. Real analytic control systems. In *IEEE Conf. on Decision and Control*, pages 5618–5623, Los Angeles, USA, Dec. 2014a. DOI: [10.1109/CDC.2014.7040268](https://doi.org/10.1109/CDC.2014.7040268)

BOOKS

- [B1] **S. Jafarpour** and A. D. Lewis. *Time-Varying Vector Fields and Their Flows*. SpringerBriefs in Mathematics. Springer International Publishing, 2014b. DOI: [10.1007/978-3-319-10139-2](https://doi.org/10.1007/978-3-319-10139-2)

INVITED TALKS

- [T23] Data-Driven Detection and Localization of Critical Transitions in Nonlinear Networks, *Workshop on Robustness, Resilience, and Early Warnings in Natural Dynamical Networks, European Control Conference (ECC 2026)*, Reykjavík, Iceland, July 2026.
- [T22] Safety Assurance in Neural Network Controlled Systems: A Mixed Monotone Approach, *ACC workshop on formal methods for systems with neural components*, July 2025 [Slides]
- [T21] Mixed Monotone Reachability in Dynamical Systems with application to safety of learning-enabled systems, *Louisiana State University, Department of Mathematics*, March 2025 (Host: Michael Malisoff) [Slides]
- [T20] Non-Euclidean Monotone Operator Theory for Robustness of Implicit Neural Networks, *Vector Institute for Artificial Intelligence, Toronto, Canada*, Feb 2025 (Host: Anastasis Kratsios) [Slides]

- [T19] Safety Assurance in Learning-enabled Autonomous Systems: A Mixed Monotone Approach *Robotic Seminar Series, University of Colorado Boulder*, Oct 2025 [[Slides](#)]
- [T18] Reachability Analysis of Dynamical Systems: A Mixed Monotone Contracting Approach, *Allerton Conference on Communication, Control, and Computing*, Sep. 2024 (Host: Daniel Liberzon) [[Slides](#)]
- [T17] Mixed-monotone Theory for Verification of Autonomous System, *Guest Lecturer in UIUC Verification of Embedded & Cyber-physical systems*, Apr. 2024 (Host: Huan Zhang) [[Slides](#)]
- [T16] Safety Assurance in Learning-enabled Autonomous Systems, *Waterloo Data and Artificial Intelligence Institute*, Mar. 2024. [[Slides](#)]
- [T15] Safety of Autonomous Systems with Learning-enabled Feedbacks, *Reliable Autonomous System Lab, Massachusetts Institute of Technology (MIT)*, Nov. 2023, (Host: Chuchu Fan) [[Slides](#)]
- [T14] Reachability Analysis of Control Systems: A Mixed Monotone Approach, *ECEE Department Seminar, University of Colorado Boulder*, Oct. 2023, [[Slides](#)]
- [T13] Interaction-aware interval reachability of neural network controlled systems, *2023 Allerton Conference on Communication, Control, and Computing*, Oct. 2023. [[Slides](#)] [[Link](#)]
- [T12] Reachability Analysis of Neural Network Controlled Systems: A Mixed Monotone Contracting Approach, *Workshop on Geometry, Topology and Control System Design, Banff Centre for Arts and Creativity, Canada*, June 2023. [[Slides](#)] [[Video](#)]
- [T11] Weak and Semi-Contraction for Large-Scale Network Systems, *LANS Seminar Talk, Argonne National Laboratory*, Apr. 2023. (Host: Adrian Maldonado) [[Slides](#)]
- [T10] Exploiting Structure in Feedback Systems with Learning-based Components, *ECEE Seminar Talk, University of Colorado Boulder*, Feb. 2023. [[Slides](#)]
- [T9] Exploiting Structure in Analysis and Design of Feedback Systems with Learning-Based Components, *Coordinated Science Laboratory, University of Illinois, Urbana Champaign (UIUC)*, Jan. 2023. (Host: Mohamed-Ali Belabbas) [[Slides](#)]
- [T8] Robustness Certificates for Implicit Neural Networks: A Mixed Monotone Contractive Approach, *Learning for Dynamics and Control (L4DC), Stanford University*, Jun. 2022. [[Slides](#)]
- [T7] Robustness of Neural Networks via Non-Euclidean Contraction Theory, *Indian Institute of Technology Delhi (virtual)*, Control Colloquium, Jun. 2022. [[Slides](#)]
- [T6] Safety and Resilience of Large-scale Networks via Contraction Theory, *University of California, Riverside, Mechanical Engineering Department*, Mar. 2022. [[Slides](#)]
- [T5] Frequency synchronization and multistability in power grids, *RSRG Virtual Seminar, California Institute of Technology*, May 2021. (Host: Steven Low) [[Slides](#)]
- [T4] Non-Euclidean Contraction and its Extensions with Applications to Network Systems, *Georgia Institute of Technology*, May 2021. (Host: Samuel Coogan) [[Slides](#)]
- [T3] Weak and Semi-Contraction for Network Systems, *Mathematical Biology Seminar, Department of Mathematics, University of Iowa*, Apr. 2021. (Host: Zahra Aminzare) [[Slides](#)]
- [T2] Stability and Control of Large-scale Nonlinear Networks, *Queen's University Control Seminar, Department of Mathematics, Queen's University*, Apr. [[Slides](#)] 2021.
- [T1] Synchronization and Multistability in Complex Networks and Power Grids, *Control Theory Seminar, Peking University*, May 2020. [[Slides](#)]

MENTORING

Students:

2024 - present	Amirreza Alavi (Ph.D. student, CU Boulder – jointly with Majid Zamani)
2026 - present	Mahadevan Seetharaman (Ph.D. student, CU Boulder – jointly with Marco Nicotra)
2026 - present	Jacob Simons (M.Sc. Student, CU Boulder)
2023 - 2026	Felipe Galarza-Jimenez (Ph.D. student, CU Boulder – jointly with Majid Zamani)
2025 - 2026	Onkar Apte (M.Sc. Student, CU Boulder)

Mentees:

2022 - 2024	Akash Harapanahalli (Ph.D. student, ECE, Georgia Tech)
2021 - 2022	Matthew Abate (Ph.D. student, ME, Georgia Tech)
2020 - 2022	Alexander Davydov (Ph.D. student, ME UCSB)
2018 - 2021	Kevin D. Smith (Ph.D. student, ECE, UCSB)
2019 - 2021	Pedro Cisneros-Velarde (Ph.D. student, ECE, UCSB)

**FUNDING &
GRANTS****Grants Awarded:**

2026	NSF-Cyber Physical Systems (total amount: \$600K, my share: \$300K) Title: <i>Reachability Analysis for Nonlinear Cyber-Physical Systems</i> . (Role: Co-PI) PI: Sriram Sankaranarayanan (University of Colorado Boulder)
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Grants Submitted:

2026	Amazon Research Award (ARA) (total amount: \$100K) Title: <i>Safe-by-Construction Learning-Based Controllers for Robotic Systems</i>
2025	NSF-Cyber Physical Systems (total amount: \$600K, my share: \$300K) Title: <i>Toward A Principled Framework for Verification and Control of Dynamical Systems under Stochastic Uncertainty</i> (Role: PI) Other PI: Yongxin Chen (Georgia Tech)
2024	NSF-Software and Hardware Foundations (total amount: \$1.2M, my share: \$210K) Title: <i>Certifiable Correctness in Neural Network Feedback Loops</i> (Role: PI) Other PIs: Samuel Coogan (Georgia Tech), Lillian Ratliff (University of Washington), Shreyas Kousik (Georgia Tech)

TEACHING**CU Boulder**

Spring 2025/26	Linear Programming (CSCI 5654)	Instructor
Fall 2025/26	Discrete Structures (CSCI 2824)	Instructor
Fall 2024/25	Advanced Linear Control (ECEN/MCEN 5448)	Instructor
Spring 2024	Feedback Control (ECEN 5138)	Instructor

UCSB

Aug. 2018	Engineering Mechanics: Dynamics (ME 16)	Instructor
Fall 2018	Nonlinear Network Systems	Guest Lecturer

Queen's University

Winter 2015	Introduction to Control Theory (MATH 332)	Instructor
Winter 2014	Lagrangian Mechanics and Control (MATH 439/836)	Instructor

**HONORS AND
AWARDS**

2024	Outstanding Reviewer, IEEE Control Systems Letters (L-CSS)
2018	Outstanding Reviewer, IEEE Control Systems Letters (L-CSS)
2011-2015	Queen's International Tuition Award, Queen's University
2011-2012	Huntly Macdonald Sinclair Tuition Fellowship, Queen's University
2011	Ranked 1 st in the M.Sc. Mechanical Engineering program, Shiraz University
2003	Awarded Silver Medal in the 23 th Iranian Student Mathematical Olympiad

Ph.D. and M.Sc. Committees:

- 2026 Felipe Galarza-Jimenez, *Ph.D. Defense Chair*, July 2026, University of Colorado Boulder.
- 2026 Rameez Wajid, *Ph.D. Defense Committee*, June 2026, University of Colorado Boulder.
- 2026 Mohammed Adib Oumer, *Ph.D. Defense Committee*, May 2026, University of Colorado Boulder.
- 2025 Sara Kamali, *Doctoral Comprehensive Exam*, University of Colorado Boulder.
- 2025 Rameez Wajid, *Doctoral Comprehensive Exam*, University of Colorado Boulder.
- 2024 Hari Krishna Hari Prasad, *Geometric nonslip contact-switching systems: gait design, analysis, and applications to tiny legged robots*, University of Colorado Boulder.
- 2024 Sai Abhishek Aravind, *Influence of Discretization on Hypersampled Model Predictive Control*, University of Colorado Boulder.

Organizer/Chair:

- 2027 Finance Chair for the “20th CPS-IoT Week”, Boulder, Colorado.
- 2024 Co-organizer of the workshop “From Formal Methods to Data-Driven Verification and Control” in 63rd IEEE Conference on Decision and Control, Milan, Italy (with Abolfazl Lavaei, Chuchu Fan, and Lars Lindemann).
- 2024 Finance Chair for the “8th IFAC Conference on Analysis and Design of Hybrid Systems” (ADHS 2024), Boulder, Colorado.
- 2017 Session Chair for “Controlled Networks and System Controllability” at the 14th SIAM Conference on Control & its Applications, Pittsburgh, Pennsylvania.

Program Committee/Editorial:

- 2025 Associate Editor handling 5 papers in 2025 IEEE International Conference on Intelligent Transportation Systems (ITSC 2025)
- 2024 Associate Editor handling 6 papers in 2024 IEEE International Conference on Intelligent Transportation Systems (ITSC 2024)

University Committee:

- 2024 Engagement & Community (EC) committee, CU Boulder Department of Electrical and Computer Engineering
- 2023 Diversity, Equity, and Inclusion (DEI) Committee, CU Boulder Department of Electrical and Computer Engineering
- 2023 Faculty and Staff Recruitment, Retention, and Retirement (FSR³) committee, CU Boulder Department of Electrical and Computer Engineering

Outreach Activity:

- 2023 Mentor for Georgia Intern-Fellowships for Teachers (GIFT)

REVIEW
ACTIVITY

- Grants □ Panelist for five NSF programs
- Journals □ Nature Communications □ IEEE Transactions on Automatic Control
□ Automatica □ SIAM Journal on Control and Optimization □ IEEE Transactions on Control of Network Systems □ IEEE Transactions on Power Systems □ IEEE Transactions on Circuits and Systems I: Regular Papers
□ IEEE Control Systems Letters □ IEEE Transactions on Control Systems Technology □ IEEE Transactions on Network Science and Engineering
□ Nonlinearity □ IEEE Transactions on Energy Conversion
- Conferences □ NeurIPS □ ICML □ ICLR □ AAAI □ IEEE Conference on Decision and Control (CDC) □ American Control Conference(ACC) □ European Control Conference (ECC)

REFERENCES

Francesco Bullo

Department of Mechanical Engineering
University of California, Santa Barbara
bullo@engineering.ucsb.edu

Sriram Sankaranarayanan

Department of Computer Science
University of Colorado Boulder
srirams@colorado.edu

Majid Zamani

Department of Computer Science
University of Colorado Boulder
majid.zamani@colorado.edu

Samuel Coogan

School of Electrical and Computer Engineering
Georgia Institute of Technology
sam.coogan@gatech.edu

Andrew D. Lewis

Department of Mathematics and Statistics
Queen's University, Canada
andrew@mast.queensu.ca